

1. Administrative & Study Identification

Full Study Title: A Phase III Multi-center, Randomized, Open-label Study to Evaluate the Efficacy and Safety of [177Lu]Lu-DOTA-TATE in Patients Newly Diagnosed With Grade 1 and Grade 2 (Ki-67 <10%) Advanced GEP-NET With High Disease Burden **Short Title/Acronym:** NETTER-3 **Posting ID:** 218041428290042074 **Protocol Number:** CAAA601A62301 **NCT Number:** NCT06784752 **Sponsor/Company Name:** Novartis Pharmaceuticals **Trial Status:** Recruiting **Investigational Product:** [177Lu]Lu-DOTA-TATE (Peptide Receptor Radionuclide Therapy) + Octreotide LAR **Phase of Development:** Phase III **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of [177Lu]Lu-DOTA-TATE plus octreotide long-acting release (LAR) versus octreotide LAR alone by assessing Progression-Free Survival (PFS) centrally assessed by a Blinded Independent Review Committee (BIRC). **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Disease Control Rate (DCR), Duration of Response (DOR), Time to Deterioration (TTD) (via EORTC QLQ-C30 and QLQ-G.I.NET21), and overall safety/ tolerability profile.

2.2 Background & Rationale To address the unmet need for effective front-line therapeutic options in newly diagnosed patients with somatostatin receptor-positive (SSTR+), well-differentiated Grade 1 and Grade 2 (Ki-67 <10%) advanced gastroenteropancreatic neuroendocrine tumors (GEP-NETs) experiencing a high disease burden. The study aims to demonstrate that combining targeted radioligand therapy with an SSA (octreotide LAR) extends progression-free survival and provides better disease control compared to standard SSA monotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Octreotide LAR alone) **Blinding:** Open-label **Randomization:** Randomized (1:1 allocation)

3.2 Planned Enrollment & Duration Targeted Enrollment: 240 participants **Number of Sites:** Multi-center (Global) **Estimated Study Start:** May 30, 2025 **Estimated Primary Completion:** January 05, 2034

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 12 years. 2. Presence of metastasized or locally advanced, unresectable (curative intent), histologically proven, well-differentiated Grade 1 or Grade 2 (Ki-67 $<10\%$) GEP-NET diagnosed within 6 months prior to screening. 3. High disease burden defined by investigator (e.g., primary/metastatic lesion > 4 cm, multiple lesions > 2 cm, elevated alkaline phosphatase $> 2.5\times$ ULN, bone/peritoneal metastasis, or symptoms due to tumor volume/hormone excess requiring active management). 4. Somatostatin receptor (SSTR) uptake on all target lesions at least as high as normal liver uptake assessed within 3 months prior to randomization. 5. Adequate bone marrow and organ function (e.g., WBC $\geq 2 \times 10^9/L$, Platelets $\geq 75 \times 10^9/L$, Hemoglobin ≥ 8 g/dL, Creatinine clearance > 40 mL/min, Total bilirubin $\leq 3 \times$ ULN). Potassium within normal limits (or up to 6.0 mmol/L acceptable). 6. ECOG performance status 0-1. 7. Presence of at least 1 measurable site of disease.

4.2 Exclusion Criteria 1. Prior administration of a therapeutic radiopharmaceutical for GEP-NET. 2. Any previous therapy with interferons, mTOR-inhibitors, chemotherapy, or other systemic therapies except somatostatin analogues (SSAs). 3. Participant who received more than 4 cycles of prior SSAs (e.g., octreotide LAR), or treatment with short-acting octreotide that cannot be interrupted for 24h, or SSAs that cannot be interrupted for 4 weeks before $[^{177}\text{Lu}]\text{Lu-DOTA-TATE}$ administration. 4. Documented RECIST v1.1 progression during previous SSA treatments for the current GEP-NET. 5. Any previous radioembolization, chemoembolization, and radiofrequency ablation for GEP-NET. 6. Known brain metastases or major surgery within 12 weeks prior to randomization. 7. Known intolerance to CT scans with intravenous (i.v.) contrast due to allergic reaction or renal insufficiency (unless participant can be imaged with MRI). 8. Hypersensitivity to any somatostatin analogues, to the Investigational Medicinal Products (IMPs) active substance, or to any of the excipients. 9. Active severe urinary incontinence, severe voiding dysfunction, or urinary obstruction requiring an indwelling/condom catheter that could prevent adhering to radiation safety instructions.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Experimental Arm:** $[^{177}\text{Lu}]\text{Lu-DOTA-TATE}$ administered 4 times during the treatment period every 8 weeks (Q8W) plus Octreotide LAR Q8W, followed by Octreotide LAR every 4 weeks (Q4W). * **Control Arm:** Octreotide LAR administered every 4 weeks (Q4W). **Route of Administration:** Intravenous (IV) for $[^{177}\text{Lu}]\text{Lu-DOTA-TATE}$; Intramuscular (IM) for Octreotide LAR. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or death.

5.2 Concomitant & Prohibited Therapy Permitted: Standard symptom management and maintenance with Somatostatin Analogues (SSAs). **Prohibited:** Other systemic anti-cancer therapies (chemotherapy, mTOR inhibitors, interferons), additional radiopharmaceuticals, or liver-directed therapies (radioembolization).

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures			
:---	:---	Screening	Up to -28 Days	Informed Consent, Biopsy Confirmation, Imaging (SSTR PET/CT), Baseline Labs	
Day 0		Baseline	Randomization, First Dose Administration, Safety Audit		
		Interim	Q8W (Lu-177 phase), then Q4W	Safety Labs, Efficacy/Tumor Assessment, Quality of Life Questionnaires, PRRT infusion	
		End of Study	Up to Jan 2034	Final Survival Assessment, PFS Event Confirmation, Exit Interview, Safety Follow-Up	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) defined as the time from randomization to the first occurrence of progression or death due to any cause. **Measurement Method:** Centrally assessed by Blinded Independent Review Committee (BIRC) according to RECIST v1.1.

7.2 Secondary & Exploratory Endpoints * Time to Deterioration (TDD) (Key Secondary) in EORTC QLQ-G.I.NET21 and EORTC QLQ-C30 domains. * Objective Response Rate (ORR) per RECIST v1.1. * Disease Control Rate (DCR) per RECIST v1.1. * Overall Survival (OS). * Duration of Response (DOR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine laboratory and clinical monitoring for hematological toxicities (lymphopenia, thrombocytopenia, anemia) and hepatotoxicity/nephrotoxicity. **Serious Adverse Events (SAEs):** Assessed according to CTCAE v5.0 criteria and reported within standard pharmacovigilance timelines. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) utilized to ensure patient safety and data integrity throughout the trial execution.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary efficacy (PFS) and survival analyses. Safety Set for AEs. **Sample Size Justification:** Target enrollment is 240 patients. Primary analysis for PFS is expected to occur after observing

approximately 88 PFS events centrally assessed by BIRC (anticipated around 33 months from study start). **Handling of Missing Data:** Standard censoring rules for time-to-event endpoints (e.g., PFS, OS) using Kaplan-Meier methodology.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring for protocol compliance, source data verification, and IMP accountability. **Audit Trail:** Validated eCRF systems, BIRC central imaging reviews, and independent audits for data clarification and locking.

11. Study Identifiers & Terminology

Primary ID: CAAA601A62301 **Registry Number:** NCT06784752 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** NETTER-3 **Official Regulatory Title:** A Phase III Multi-center, Randomized, Open-label Study to Evaluate the Efficacy and Safety of [177Lu]Lu-DOTA-TATE in Patients Newly Diagnosed With Grade 1 and Grade 2 (Ki-67 <10%) Advanced GEP-NET With High Disease Burden

Disease/Condition Terminology: * **Disease Areas:** Somatostatin Receptor Positive (SSTR+), Gastroenteropancreatic Neuroendocrine Tumor (GEP-NET) * **Preferred Name:** Neuroendocrine Tumors * **Preferred UMLS Name:** Neuroendocrine Tumor, Somatostatin Receptor Positive * **Definition:** Tumors whose cells possess secretory granules and originate from the neuroectoderm, i.e., the cells of the ectoblast or epiblast that program the neuroendocrine system. Common properties across most neuroendocrine tumors include ectopic hormone production (often via APUD CELLS), the presence of tumor-associated antigens, and isozyme composition. * **Semantic Type:** Neoplastic Process * **Classification Codes:** * **MeSH Code:** D018358 * **HPO Code:** HP:0100634 * **SNOMED ID:** 1288045008 * **SNOMED Hierarchy:** 722688002 | Malignant epithelial neoplasm (disorder) > 1290125003 | Neuroendocrine neoplasm, malignant (disorder) > 64572001 | Disease (disorder)

Drug Terminology: * **Drug Name / Preferred Name:** octreotide * **ATC Code:** H01CB02 * **Trade Names:** Sandostatin, Sandostatin LAR, Bynfezia, Mycapssa * **EMA URL:** [Mycapssa EPAR Product Information](#)

12. Clinical Framework (GEP-NET Specific)

Primary Condition: Advanced Gastroenteropancreatic Neuroendocrine Tumors (GEP-NET) **Diagnosis Threshold:** Grade 1 and Grade 2 (Ki-67 <10%), High Disease Burden, SSTR+ **Medical Methodology:** Peptide Receptor Radionuclide Therapy (PRRT) combined with SSA

Optimization Target: Delay in tumor progression (PFS) and objective response/symptom control

13. Study Procedures & Methodology

Management Pathway: Experimental PRRT ([177Lu]Lu-DOTA-TATE) + Standard SSA clinical pathway **Education Protocol (HCP):** Specialized site training on radiopharmaceutical administration, radiation safety, and dosimetry. **Education Protocol (Patient):** Radiation safety precautions post-infusion and symptom management tracking. **Quality Audit Mechanism:** Blinded Independent Review Committee (BIRC) for unbiased radiological assessment (RECIST v1.1).

14. Observation & Follow-Up Schedule

Total Duration: Up to 104 months (Study completion estimated Jan 2034) **Onsite Visit Cadence:** Every 8 weeks (Q8W) during experimental treatment, transitioning to Every 4 weeks (Q4W) for maintenance **Remote Monitoring Frequency:** Routine AE, safety, and survival follow-up per protocol schedules. **Checklist Review Interval:** Regular tumor assessments (imaging) synchronized to trial milestones (e.g., Q8W to Q12W) to capture PFS events.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				